

CHEN YONGWEI

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SENIOR SOFTWARE ENGINEER

Software Engineer with 5 years of experience in large-scale distributed systems. Proven track record of designing and developing large-scale Music Information Retrieval system, searching through millions of songs within milliseconds. Proficient in C++, Go, Python and adept at optimizing system infrastructure for high performance and cost-savings. Strong expertise in high-performance computing techniques and inference model optimization.

WORK EXPERIENCE

TikTok / ByteDance

Senior Software Engineer | C++, Python

Singapore

Dec 2023 – Present

- Developed large-scale distributed Music Information Retrieval systems with millions of songs and thousands of QPS.
- Designed extensive cloud-based audio fingerprinting cluster, searching through millions of songs within milliseconds.
- Optimized RPC overheads through resolving bottlenecks, doubling overall QPS, achieving 100% cost-saving.
- Deployed and maintained large-scale physical machine-based MIR clusters, e.g. Sapphire Rapids/Ice Lake.
- Architected distributed C++ model inference engine for Universal Sound Separation, handling hundreds of QPS.
- Orchestrated 0→1 development of the CPU-based C++ OpenVINO inference engine across a team of 3.
- Designed distributed inference components to overcome inference latency bottlenecks, minimising latency by 4x.
- Optimized inference efficiency through model modification and pruning, increasing QPS by 2x with same precision level.
- Researched high-performance computing techniques and model inference optimization, increasing QPS up to 10x.
- Experimented performance optimization techniques for CPU-bound and memory-bound systems, e.g. SIMD, NUMA.
- Validated and integrated performance optimization across multiple services, achieving overall cost-efficiency by 3x.
- Developed C++ STFT/ISTFT library for audio signal processing with SIMD acceleration, reducing latency by 20%.

uWave

Software Engineer | Go, Kubernetes

Singapore

Jul 2020 – Nov 2023

- Led the development of over 30 backend microservices for app-based platform, serving more than 40,000 users.
- Developed every microservice from the ground up with extensive CI/CD pipeline to Kubernetes using Helm Charts.
- Architected end-to-end complex features across multiple services, e.g. authorization, content personalization.
- Maintained the Kubernetes microservice cluster with New Relic metrics integration and alerts, achieving 99.999% SLA.
- Optimized infrastructure costs through Java→Go code refactoring, CDN-egress optimization.

MOH Office for Healthcare Transformation

Software Engineer Intern | Swift, iOS, Java, Android

Singapore

May 2020 – Jul 2020

- Developed the digital phenotyping platform for iOS, allowing the onboarding of iPhone users, scaling up to 10,000 users.
- Simplified onboarding process using QR codes for both iOS and Android mobile devices.

EDUCATION

Nanyang Technological University

Bachelor of Engineering (Computer Science) and Bachelor of Business (Business Analytics)

Singapore

Meridian Junior College

GCE 'A' Levels - **6 Distinctions**

Singapore

AWARDS AND ACHIEVEMENTS

- 2020 - Aon x NTU Data Hackathon:** 1st place out of 7 teams, utilising real-life data for customer matching
- 2019 - Shopee National Data Science Challenge:** 1st place out of 360 teams, classifying online listings
- 2018 - iNTUition v5.0 Hackathon:** Best AI Hack, innovating Generative AI solution for audio creativity
- 2018 - Bambu Fintech Challenge:** 1st place, designing and developing web applications for financial planning

SKILLS

Experience: C++, Go, Python, numactl, SIMD, TiKV, MinIO, Pytorch, OpenVINO, ONNX, bRPC

Familiar: Java, React, Next.js, Vue, Flutter, Tensorflow, SQL, NoSQL, MongoDB, S3, Redis, Kubernetes, Microservices

Languages: English (Native), Mandarin Chinese (Professional Proficiency)